

Ways and means: Inks, yarns and embroidery, too

To bring technology into your clothes, you need some way of weaving the electronics into mainstream materials, because nobody likes to wear a plastic dress. Here are some of the ways in which this can be achieved:

Touch of silver turns fabrics into electronics. DuPont is one of the well-known manufacturers of stretchable electronic materials like conductors, encapsulants and sensors, made with silver and dielectric inks. You can use these manufacturing-ready materials to make thin, form-fitting circuits that can be easily fused into any kind of fabric. These electronic

materials are supposed to withstand around 100 wash cycles. The inks are printed using an inkjet printer, which can manage complex multi-track circuits as well as entire fabric coverage. So your garment itself becomes a printed circuit board (PCB).

Graphene based inks. According to a report published in the journal *Carbon*, a team of scientists at University of Cambridge in the UK and Jiangnan University in China has developed a method of making cotton textiles conductive by impregnating these with graphene based ink. Cotton is one of the most popular, durable and comfortable materials used for making clothing, and should

A new generation of smart fabrics

Smart fabrics might not seem so novel to you if you just think of these as garments with a chip attached to these. Clipping devices onto one's dress, or even stitching a chip into it, is old news. We are beginning to see a new generation of clothing, woven with fibres that themselves are sensors and computers.

materials are supposedly durable and stretchable, and can be washed along with the fabric in a regular washing machine using normal detergents. DuPont claims that these can withstand around 100 wash cycles.

Other technologies such as Noble Biomaterials' Circuitex also use silver for making fabrics conductive. Circuitex works by bonding a non-conductive polymer like nylon with 99.99 per cent pure silver. The silver covers the whole surface around the fibre, which creates a highly conductive and uniform material that can be processed in traditional textile manufacturing machinery. Circuitex technology is used in products like Adidas miCoach and Ralph Lauren's PoloTech Shirt.

The National Physical Laboratory of the UK takes a similar approach to integrating electronics into any type of clothing, including cotton fabrics. Their method begins by attaching a nano-metal seed layer to the fibres. This acts as a catalyst towards a secondary electro-less metal-plating process, which encapsulates the fibres. The resulting textile is very conductive, flexible and

ideally be used for e-textiles, too, if we want any kind of mass adoption.

To achieve this, the team has developed inks of chemically-modified graphene flakes that are more adhesive to cotton fibres than unmodified graphene. The ink is deposited on the fabric and then subjected to heat treatment for improving its conductivity. The modified graphene adheres to the cotton just like any other colour dye, which ensures that the fabric remains conductive even after several washes. The team demonstrated the effectiveness of the technique using a motion sensor based on the conductive cotton.

This method is expected to become quite popular in the future because graphene ink is chemically-compatible with cotton, unlike certain methods that require a polymer or plastic base. Also, it is less expensive and environment-friendly compared to metal based conductive inks, especially those using pure silver.

Techie yarn. Project Jacquard by Google proposes another method of weaving electronics into garments.

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